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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2019

Code: 211405 | ***Subject: Discrete Mathematical Structure and Graph Theory*** | ***Time: 03 Hours*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer (any seven): **[2x7=14]**

(a) The statement $p \rightarrow q$ is logically equivalent to
(i) $p$vee$q$ (iii) sim$p$vee$q$
(ii) $p$veesim$q$ (iv) $p \rightarrow q$

(b) The contrapositive of the conditional statement $p \rightarrow q$ is
(i) $q \rightarrow p$ (iii) $p \rightarrow q$
(ii) sim$p \rightarrow$ sim$q$ (iv) sim$q \rightarrow$ sim$p$

(c) If A and B are two nonempty sets having n elements in common, then A×B and B×A will have how many elements in common?
(i) 2^n (iii) n^4
(ii) n^2 (iv) 2n

(d) If a set A have n elements, then how many relations will be there on set A?
(i) n^2 (iii) 2^n
(ii) 2^{n^2} (iv) 2n

(e) If $P(\varphi)$ represents the power set of φ, then $n(P(P(P(\varphi))))$ equal to
(i) 1 (iii) 3
(ii) 2 (iv) 4

(f) The number of edges in a bipartite graph with n vertices is at most
(i) $n^2/2$ (iii) n^2
(ii) $n^2/4$ (iv) 2n

(g) If (S, *) is a monoid, where S ={1, 2, 3, 6} is defined by $a * b = lcm(a, b)$, and where a, $b$in$S$, then the identity element is
(i) 1 (iii) 3
(ii) 2 (iv) 6

(h) The total number of subgroups of a group G of prime order is
(i) 1 (iii) 3
(ii) 2 (iv) 4

(i) For the poset {3, 5, 9, 15, 24, 45}; divisor of the glb of {3, 5} is
(i) 3 (iii) 15
(ii) 5 (iv) 45

(j) The number of pendant vertices of a full-binary tree is
(i) [n+1] / [2] (iii) [2n+1] / [2]
(ii) [n−1] / [2] (iv) [2n−1] / [2]

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Q.2 Solve both questions : **[14]**

(a) Using truth table, show that-
(i) $((p \rightarrow (q \rightarrow r))) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$ is a tautology.
(ii) sim$(q \rightarrow r)$wedger wedge$(p \rightarrow q)$ is a contradiction.

(b) Obtain the principal disjunctive normal form (PDNF) and principal conjunctive normal form (PCNF) of the statement $(p \rightarrow (q$wedger$))$wedge(sim$p \rightarrow$ (sim$q$wedgesim$r$)).

Q.3 Solve both questions : **[7+7=14]**

(a) For any sets A and B, prove that
(i) (Acup$B$)' = A'cap$B$';
(ii) (Acap$B$)' = A'cup$B$'.

(b) If two sets A and B have n elements in common, then show that the sets A×B and B×A will have 2^n elements in common.

Q.4 Solve both questions : **[14]**

(a) If R is the relation on the set of positive integers, such that (a, b)in$R$ if and only if $a^2 + b$ is even, prove that R is an equivalence relation.

(b) Define partition of a set. If the relation R on the set of integers Z is defined by a R b iff $a$cong$b$pmod4, find the partition induced by R.

Q.5 Solve both questions : **[7+7=14]**

(a) If R and S be relations on A ={1, 2, 3} represented by the matrices $M_R = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix}$ and $M_S = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$.
Find the matrices that represent (i) Rcup$S$, (ii) Rcap$S$, (iii) $R \circ S$, (iv) $R - S$, (v) R', (vi) $R \circ R$ and (vii) Roplus$S$.

(b) Draw the Hasse diagram representing the partial ordering {(A,B) : AsubseteqB} on the power set $P(S)$ where $S = \{a,b,c\}$. Find the maximal, minimal, greatest and least elements of the poset.

Q.6 Solve both questions : **[7+7=14]**

(a) Define characteristic function of a set. If A and B are any two subsets of universal set U, then show that:
$f_{A \text{cup} B}(x) = f_A(x) + f_B(x) - f_{A \text{cap} B}(x)$, for all $x$in$U$

(b) If functions f, g, h: $Z \rightarrow Z$ are defined as $f(x) = x - 1$, g$(x) = 3x$ and h$(x) = 0$, &*if x is even*
1, &*if x is odd*
Verify that $f \circ (g \circ h) = (f \circ g) \circ h$.

Q.7 Solve both questions : **[7+7=14]**

(a) Show that every group of order 3 is cyclic.

(b) Prove that the necessary and sufficient condition for a non-empty set H of a group (G, *) to be a subgroup is a, $b$inH$\Rightarrow a * b^{-1}$inH.

Q.8 Solve both questions : **[7+7=14]**

(a) Show that the order of a subgroup of a finite group is a divisor of the order of the group.

(b) Prove that the set S of all real numbers of the form $a + b\sqrt{(2)}$ where a, b are integers is an integral domain with respect to usual addition and multiplication.

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Q.9 Solve both questions : **[7+7=14]**

(a) Define adjacency matrix and incidence matrix of graph G. Draw the graph represented by the adjacency matrix

$$\begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$$

(b) Show that a tree with n vertices has $(n-1)$ edges.

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